

Two Horizons: Adversarial Technical Review

Review date: September 12, 2026. Subject: Jason Shroyer’s “Two Horizons: Coarse-Graining Defects, Fluid Computation, and Regularity Budgets,” version 0.1, and accompanying review commission.

Prepared by Perplexity Computer. This is an AI-assisted mathematical and literature review, not an independent human expert endorsement or a replayed Lean audit.

Verdicts

Question	Verdict	Consequence
Q1: flux-controlled projected Galerkin error	False as written; corrected residual-based estimation is known.	Change the residual, comparison trajectory, and novelty claim.
Q2: computation requires zero flux	Not established; ill-posed without domain, metric, forcing, and coding constraints.	Remove the necessity claim. The precise three-dimensional nonzero-flux realization is not settled by this review.
Q3: a monotone separator for the proposed three phases	The phase diagnostic is incompatible with bounded energy; no general separator was identified.	Use a precisely defined existing regularity criterion, not divergence at every fixed cutoff.

The distinction between approximation reliability and continuation remains useful. The identifications “defect = flux,” “computation requires zero physical-scale defect,” and “finite-energy blowup makes every fixed-cutoff defect diverge” do not follow from that distinction. These findings do not depend on accepting or rejecting the September 2026 blowup announcements.

Review scope and evidence boundary

The attached 526-line manuscript was read in full. The mathematical findings below concern its stated definitions and claims, independently of whether their associated Lean declarations compile. Calculations explicitly presented as derivations are mathematical arguments supplied by this review, not newly kernel-checked results.

The requested repository audit could not be completed. A read-only clone of JasonShroyer/sgc-lean, branch opus/two-horizons, returned Remote branch opus/two-horizons not found in upstream origin; both the GitHub branch API and git ls-remote --heads omitted that branch. The accessible repository was reported as public, rather than the private review repository described in the commission, and its available default tree did not contain the named headline modules. This establishes an access/ref mismatch, not that the local development does not exist.

Consequently, the 48-file closure, 170-theorem count, 38 supporting axioms, receipt hashes, exceptional TrajectoryClosure dependencies, and clean-build report remain user-reported rather than independently audited here. No lakefile, Lean macro, project executable, or build

command was run, and no repository was modified. An archive or accessible immutable commit is needed to finish that part of the commission.

The decisive mathematical distinction

Work on the flat periodic torus $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$, with smooth divergence-free fields and the L^2 inner product. Let P_N be the orthogonal Fourier cutoff $|k| \leq N$, let $\mathbb{P}$ be the Leray projector, and write

$$B(a, b) = \mathbb{P}((a \cdot \nabla)b), \quad A = -\Delta.$$

The equation is $u_t + \nu Au + B(u, u) = f$, with projected forcing understood. Set $v = P_N u$ and $w = (I - P_N)u$. The commission's positive-sign convention gives

$$D_N(u) = (I - P_N)B(v, v).$$

The manuscript sometimes uses the opposite sign because it defines the defect from the full evolution vector field. A global sign change does not repair the distinctions below.

Object	Definition	What it measures
Outgoing nonlinear defect	$D_N(u) = (I - P_N)B(v, v)$	High modes generated by low-low interaction
Resolved closure residual	$R_N(u) = P_N[B(u, u) - B(v, v)]$	Unresolved feedback missing from the autonomous low-mode equation
Lifted Galerkin residual	$r_N = (I - P_N)[B(u_N, u_N) - f]$	Failure of the numerical trajectory, embedded in the full space, to solve the full equation
Signed forward energy flux	$\Pi_N(u) = \langle v, B(u, u) \rangle$	Nonlinear loss rate of resolved kinetic energy

These are not interchangeable. In particular, D_N takes values in the high-mode subspace, R_N in the low-mode subspace, and Π_N is a signed scalar.

The exact projected error equation

Let u_N solve the autonomous Galerkin equation with initial datum $P_N u(0)$:

$$\dot{u}_N + \nu Au_N + P_N B(u_N, u_N) = P_N f.$$

Then $e = P_N u - u_N$ satisfies the exact identity

$$\dot{e} + \nu Ae + P_N[B(P_N u, P_N u) - B(u_N, u_N)] = -R_N(u).$$

This follows by projecting the full equation and subtracting the Galerkin equation. The forcing term is R_N , not D_N ; pressure is not the obstruction, since the Leray projection has already removed it. Expanding $u = v + w$ gives

$$R_N = P_N[B(v, w) + B(w, v) + B(w, w)].$$

The finite-state invariant-subspace theorem quoted in the paper cannot simply be relabelled as this nonlinear projected-error theorem. Its comparison of full evolution from coarse initial data with an embedded coarse evolution has a different orientation.

The exact energy-flux identity

The low-mode energy balance is

$$\frac{d}{dt} \frac{1}{2} \|v\|_2^2 + \nu \|\nabla v\|_2^2 = \langle v, f \rangle - \Pi_N(u).$$

Integration by parts for divergence-free fields yields

$$\boxed{\Pi_N(u) = \langle v, R_N(u) \rangle = -\langle w, D_N(u) \rangle + \langle v, B(w, w) \rangle.}$$

Thus pairing the proposed defect with the fine field captures only one contribution, with a convention-dependent sign; it omits the high-high contribution to resolved transfer. Moreover, a signed scalar flux can vanish through orthogonality or cancellation while a vector residual is nonzero. An acceleration residual and an energy-transfer rate also have different physical dimensions before normalization.

The identity $D_N = (I - P_N)F(P_N u)$ in manuscript Section 6.1 additionally drops unresolved forcing. Since $(I - P_N)\Delta P_N u = 0$, the full expression is

$$(I - P_N)F(t, P_N u) = -D_N(u) + (I - P_N)f(t).$$

The displayed manuscript equality therefore requires an explicit restriction on the forcing or a definition confined to the nonlinear component.

Counterexamples and a no-go estimate

A smooth unforced counterexample to Q1

Take cutoff $N = 1$ and the two-dimensional initial field, embedded in three dimensions,

$$u_0(x, y, z) = (\cos(-2x + y), \cos(2x) + 2\cos(-2x + y), 0).$$

Its divergence is zero. Its only initial wavevectors are $\pm(2, 0, 0)$ and $\pm(-2, 1, 0)$, all outside the cutoff, so $P_1 u_0 = 0$. Direct Fourier multiplication and Leray projection give

$$P_1 B(u_0, u_0) = -\frac{1}{2}(\sin y, 0, 0).$$

For either unforced Euler or positive-viscosity Navier-Stokes, on any interval of smooth existence,

$$\left. \frac{d}{dt} P_1 u(t) \right|_{t=0} = \frac{1}{2}(\sin y, 0, 0).$$

The Galerkin solution with zero initial data is identically zero. Nevertheless, $D_1(u(t)) = 0$ throughout the smooth interval: the solution remains independent of z , has zero mean and zero vertical velocity, and preserves its π -periodicity in x . Its only possible retained nonzero modes

are $(0, \pm 1, 0)$; incompressibility therefore makes $P_1 u$ a shear $(h(y, t), 0, 0)$, whose self-advection vanishes.

Consequently,

$$\|P_1 u(t) - u_1(t)\|_2 > 0 \quad \text{for sufficiently small } t > 0, \quad \int_0^t \|D_1(u(s))\|_2 e^{C(t-s)} ds = 0.$$

This is an explicit counterexample to Q1 as written, without blowup, forcing, a delicate pressure estimate, or a limiting argument. It already works when the “fine” system is the finite $M = 3$ Galerkin system and the coarse system has $N = 1$: both initial waves are retained by P_3 , and the same symmetries persist. That gives a finite-dimensional falsification test before any continuum formalization. The Fourier coefficients were also checked using exact rational arithmetic: at wavevector $(0, 1, 0)$, $B(u_0, u_0)$ has coefficient $i(1/4, 0, 0)$.

A nonzero defect with zero scalar flux

For cutoff $N = 2$, take

$$v(x, y, z) = (0, \cos(2x), \cos y).$$

This divergence-free field is fully retained, while

$$B(v, v) = (0, 0, -\cos(2x) \sin y)$$

has only wavevectors of length $\sqrt{5} > 2$. It is already divergence-free, so $D_2(v) = B(v, v) \neq 0$, but

$$\langle v, D_2(v) \rangle = \Pi_2(v) = 0.$$

Any proposed two-sided comparison between the defect norm and the absolute scalar energy flux fails on this example. Calling nonzero D_N “nonzero energy flux” would erase exactly the distinction the example demonstrates.

Bounded energy prevents fixed-cutoff defect divergence

For every fixed $N \geq 1$, the elementary band-limited estimates give

$$\begin{aligned} \|D_N(u)\|_2 &\leq \|(P_N u \cdot \nabla) P_N u\|_2 \\ &\leq \|P_N u\|_\infty \|\nabla P_N u\|_2 \\ &\leq C N^{5/2} \|u\|_2^2. \end{aligned}$$

Here C depends only on the torus normalization. The $N^{3/2}$ factor comes from Cauchy–Schwarz over the retained Fourier coefficients and the N factor from differentiation. Therefore,

$$\sup_{t < T_*} \|u(t)\|_2 < \infty \quad \implies \quad \sup_{t < T_*} \|D_N(u(t))\|_2 < \infty \quad \text{for every fixed } N.$$

The same conclusion applies to any fixed Sobolev norm of D_N , since its Fourier support lies below $2N$. Also,

$$\|P_N B(u, u)\|_2 \leq CN^{5/2} \|u\|_2^2,$$

because each Fourier coefficient of $u \otimes u$ is bounded by $C\|u\|_2^2$, and summing squared differentiated coefficients over $|k| \leq N$ gives the stated factor. Thus even the corrected low-mode residual remains bounded at fixed N under a uniform energy bound.

This directly invalidates the proposed phase diagnostic for a finite-energy singularity. The logical caveat matters: the estimate does not prove that an unforced singularity exists, and it is not itself a counterexample to a potentially vacuous unforced implication. It proves that any finite-energy singularity, if one exists, cannot have the behavior required by Conjecture 6.2. If that conjecture were proved for all unforced solutions, it would exclude such singularities altogether rather than describe their mechanism.

The issue is the order of quantifiers. Loss of regularity can concern $N \rightarrow \infty$, a moving cutoff $N(t)$, or failure of a bound uniform in N , while every fixed finite observation remains bounded. Replacing “every fixed N ” by “some moving $N(t)$ ” removes this elementary obstruction but does not automatically produce a true regularity criterion.

An obstruction to a universal Euler Lyapunov function

There is a useful direct constraint on Q3. Suppose an instantaneous functional $H(u, N)$, at fixed N , is even under velocity reversal, $H(-u, N) = H(u, N)$, and is nonincreasing along every smooth unforced Euler solution. For a solution segment $u(t)$, $0 \leq t \leq T$, the time-reversed field $\tilde{u}(t) = -u(T - t)$ is another Euler solution. Monotonicity applied to both segments forces $H(u(t), N)$ to be constant.

Thus a nontrivial, sign-even, universally decreasing instantaneous “free energy” is unavailable for reversible Euler dynamics. This argument does not apply to viscous Navier–Stokes, accumulated time-integrals, an irreversible rule for changing the cutoff, or a restricted class of trajectories. It does explain why importing an irreversible finite-state entropy picture directly into Euler is structurally problematic.

A corrected first L1 theorem

The appropriate starting point is an inhomogeneous residual estimate. Let $v(t)$ be a smooth divergence-free approximate solution and define its full residual by

$$r = \dot{v} + \nu A v + B(v, v) - f.$$

For a smooth exact solution u , put $z = u - v$. Divergence-free energy cancellation gives

$$\frac{1}{2} \frac{d}{dt} \|z\|_2^2 + \nu \|\nabla z\|_2^2 \leq \|\nabla v\|_\infty \|z\|_2^2 + \|r\|_2 \|z\|_2.$$

Writing $a(t) = \|\nabla v(t)\|_\infty$, the resulting comparison estimate is

$$\|u(t) - v(t)\|_2 \leq e^{\int_0^t a(\tau) d\tau} \|u(0) - v(0)\|_2 + \int_0^t e^{\int_s^t a(\tau) d\tau} \|r(s)\|_2 ds.$$

The norm inequality at zeros is handled by regularization or an upper-Dini-derivative argument. For a lifted Galerkin solution, $r = (I - P_N)[B(u_N, u_N) - f]$. This is an actual a posteriori quantity when the approximate trajectory and forcing are available, unlike a residual requiring the unknown exact solution. The full-error estimate also bounds projected error by orthogonal contraction, but its initial term includes the unresolved initial tail.

This is not a literal instantiation of the quoted homogeneous L0 hypothesis $\|x'\| \leq W\|x\|$. A nonzero residual can generate error from zero initial error, which a homogeneous inequality would prohibit; moreover, energy cancellation controls radial norm growth without requiring the full derivative norm to satisfy the same bound. The next Lean result should therefore be an inhomogeneous norm/energy comparison theorem, followed by its finite-dimensional fluid instantiation.

A bounded first implementation should compare M -mode and N -mode systems with $M > N$, on the same explicit Fourier state space. It should prove the residual identity, trilinear cancellation, and error estimate before any continuum claim. Constants depending on N , M , or the approximate trajectory should remain visible rather than being silently described as uniform.

Novelty assessment

The residual-controlled a posteriori program is established prior art, not merely a resemblance to classical Sobolev convergence bounds: Morosi and Pizzocchero explicitly compute the Galerkin differential error in equation (6.20) as the unresolved nonlinear term minus unresolved forcing, and use control inequalities to bound the approximation error and certify an existence interval (Morosi–Pizzocchero, equations 3.10, 4.24–4.27, and 6.20). Their Section 7 supplies a Fourier-computable Sobolev residual estimator, so the specific idea “use the spectral defect as the currency of an a posteriori estimate” already has a direct precedent (Morosi–Pizzocchero, equations 7.12–7.15).

An earlier robustness and numerical-verification framework also turns approximate-solution residuals into strong-solution guarantees under explicit Sobolev smallness conditions (Chernyshenko–Constantin–Robinson–Titi, Theorem 3 and Corollary 5). These references rule out a broad novelty claim for the repaired idea; they do not establish that a proposed Lean implementation, sharper computable constants, or an SGC-specific observable theorem already exists.

The elementary L^2 estimate derived above is conditional on a sufficiently regular exact solution over the comparison interval. It must not be promoted, without a separate strong-norm/bootstrap argument, into an existence or regularity certificate. The cited Sobolev-control frameworks address that stronger task with additional hypotheses and inequalities.

Q2: computation and transfer

The known constructions do not imply a zero-flux necessity theorem. The PNAS construction realizes a Turing-complete area-preserving disk map through a steady Euler carrier on a three-sphere with an adapted metric; its damped Navier–Stokes variant is explicitly limited to a finite simulated-time budget and does not establish eternal Navier–Stokes computation (Cardona–

Miranda–Peralta-Salas–Presas, Theorems 5.2 and 6.1 and the Navier–Stokes discussion). Separately, time-dependent Eulerian Turing universality has been constructed on a high-dimensional compact Riemannian manifold, so “every known realization is stationary” is false unless the statement is explicitly restricted to the relevant three-dimensional class (Cardona–Miranda–Peralta-Salas, abstract).

The geometry restrictions are material. The Euclidean Beltrami paper establishes full universality on $\mathbb{R}^3$ with infinite energy, while its flat-torus construction simulates tape-bounded machines and explicitly leaves a Turing-complete flat-torus Beltrami field unresolved (Computability and Beltrami fields in Euclidean space). The stationary viscous construction requires a nowhere-vanishing harmonic carrier and generally a deformation of the metric, with a Hodge-Laplacian viscosity operator; $H^1 \neq 0$ alone is not the stated sufficient hypothesis (Dyhr–González-Prieto–Miranda–Peralta-Salas, Theorem A).

The concrete three-dimensional question remains unresolved by the evidence reviewed: no construction or impossibility theorem was verified here for the specified combination of an exact Moore return map, fixed flat geometry, finite energy, and nonzero physical spectral transfer at infinitely many cutoffs. “Nonzero D_N ” is also weaker than “nonzero signed flux,” and neither condition quantifies scale-nonlocal transfer; those requirements need separate definitions. A zero symbolic closure defect is a property of a chosen coding and dynamics, not an invariant statement about every Fourier truncation of a fluid realization.

Forcing must be fixed before treating Q2 as a PDE obstruction. If arbitrary smooth forcing may be chosen after the velocity field, any prescribed smooth divergence-free u satisfies forced Navier–Stokes after setting

$$f = u_t + \nu Au + B(u, u).$$

Likewise, composing a carrier flow with a smooth time-periodic volume-preserving coordinate loop can preserve its stroboscopic map while making its Eulerian velocity time-dependent; residual forcing enforces the PDE. This is a kinematic mechanism, not a verified solution of the full flat-torus, infinitely-many-cutoffs request, and it must not be advertised as computational blowup. A useful sharpened question should fix the forcing class, domain/metric, observation map, exact-versus-robust simulation requirement, and actual flux condition before asking for an obstruction.

An explicit stationary forced field with positive flux at every cutoff

The implication “stationary, hence zero energy flux across every scale” is false for general forced flows. Here is a direct analytic example, using normalized spatial averages in the inner product. Let

$$a(x) = \sum_{m \geq 1} e^{-m^2} \cos(mx), \quad V = (0, a(x), \cos y + \eta a(x) \sin y), \quad \eta > 0.$$

The field is smooth, divergence-free, and finite-energy. Writing $A_N = \sum_{m=1}^N e^{-m^2} \cos(mx)$ and $A_0 = 0$, for integer cutoffs $N \geq 1$,

$$P_N V = (0, A_N, \cos y + \eta A_{N-1} \sin y), \quad B(V, V) = (-a \sin y + \eta a^2 \cos y) e_z.$$

Direct orthogonality gives

$$\Pi_N(V) = \frac{\eta}{4} \sum_{m \geq N} e^{-2m^2} > 0, \quad \|D_N(V)\|_2 \geq \frac{1}{2} e^{-N^2} > 0.$$

Taking the time-independent force $f = \nu AV + B(V, V)$ makes V an exact stationary forced Navier–Stokes solution. Setting $\nu = 0$ gives the corresponding forced Euler example. This does not make V a computer, but it disproves the stationarity-to-zero-flux inference and shows that nonzero flux at infinitely many cutoffs need not indicate singular behavior. In this example $\Pi_N \rightarrow 0$ rapidly.

There is also a precise conditional coexistence mechanism. Suppose a smooth divergence-free carrier X on this same flat torus has a time- τ map realizing the intended machine on a coded set. Define $Y = V - X$, $h(t) = \frac{\tau}{2\pi} \sin(2\pi t/\tau)$, and $g_t = \phi_Y^{h(t)}$, where ϕ_Y denotes the flow of Y . Then

$$u(t) = h'(t)Y + (g_t)_*X, \quad \Psi_t = g_t \circ \phi_X^t, \quad \Psi_\tau = \phi_X^\tau, \quad u(k\tau) = V.$$

This is a smooth divergence-free time-periodic velocity, realized by residual forcing, with the same stroboscopic computation and positive flux at every integer cutoff at recurrent phases. For each fixed N , positivity also persists locally in time by continuity, but no common neighborhood for all N is claimed. The construction is conditional on the carrier's realization in the same geometry; it does not establish that missing hypothesis, preserve arbitrary spatial first-return maps, give an unforced example, or produce blowup.

Q3: which wavenumber, and which budget?

No universal signed-flux Lyapunov function separating regularity from singularity was identified in the inspected primary literature. There are precise regularity and determining-mode criteria, but they are not a monotone three-phase classification, and they are not equivalent to the paper's D_N . The Euler time-reversal obstruction and fixed-cutoff energy bound above give concrete reasons to abandon the stated phase picture rather than merely search for a missing monotone.

The dissipation wavenumber

Cheskidov and Shvydkoy define a dyadic dissipation threshold by

$$\Lambda_{\text{dis}}(t) = \min\{\lambda_q : \lambda_p^{-1} \|u_p(t)\|_\infty < c_0 \nu \text{ for every } p > q\}, \quad \lambda_q = 2^q.$$

Their theorem, stated for unforced Navier–Stokes on $\mathbb{R}^3$, says that a Leray–Hopf solution on $[0, T]$ is regular on $(0, T]$ if its low-mode vorticity budget is finite; writing $\lambda_{Q(t)} = \Lambda_{\text{dis}}(t)$, the density is

$$b_rmCS(t) = |u_leQ(t)(t)|_{B_{inftyfty, infty}^1} \sim |\omega_leQ(t)(t)|_{B_{inftyfty, infty}^0}.$$

The viscosity, spatial domain, solution class, and dyadic definitions belong to the theorem, not to optional notation ([Cheskidov–Shvydkoy, Theorems 1.1 and 3.2](#)).

The same paper gives the unconditional sufficient condition $\Lambda_{\text{dis}} \in L^{5/2}(0, T)$, and gives $\Lambda_{\text{dis}} \in L^2(0, T)$ as sufficient with the additional hypothesis $u \in L^\infty(0, T; B_{\infty, \infty}^{-1})$ ([Cheskidov-Shvydkoy, Theorem 3.2 and Corollary 3.4](#)). Thus the manuscript's bare attribution "their regularity criterion $\int \Lambda^2 dt < \infty$ " omits a substantive hypothesis for this definition; it is not the theorem as stated in that source.

Determining wavenumbers are different objects

The determining-mode construction of Cheskidov, Dai, and Kavlie adds low-mode control and proves convergence or uniqueness from agreement of low modes under its stated time-history hypotheses ([Cheskidov-Dai-Kavlie, definition of \$\Lambda_{u,r}\$, Theorems 1.1-1.2](#)). Agreement of observations determining a solution is not the same assertion as an autonomous Galerkin trajectory approximating those observations.

A later Cheskidov-Dai definition includes the explicit condition

alongside a weighted high-mode condition ([Cheskidov-Dai, definition of \$\Lambda_u\$](#)). This makes a $\nu \int \Lambda_u^2 dt$ budget a plausible derived target for that stronger cutoff, but a formal implication must check compatibility with the regularity criterion's cutoff, constants, domain, and forcing hypotheses; the cited paper does not itself state the proposed integral criterion.

The safe L2 target is therefore the named low-mode regularity density b_{CS} , or a fully specified determining-cutoff variant with a proved comparison to that density. A future SGC estimate should relate an actual residual or observable to that target, not identify the quantities by terminology. Neither unboundedness of a wavenumber nor divergence of its integral automatically means a full limit $\Lambda(t) \rightarrow \infty$, monotonicity in time, or loss of every fixed coarse description.

External-artifact status and its limited role

The external repository states forced Navier-Stokes alternatives C/D and an unforced Euler singularity claim ([NavierStokesAndEuler README](#)). The official Clay problem statement includes C/D among its designated alternatives, so leaving A/B untouched must not be confused with showing that a valid C/D proof would be irrelevant to the Millennium formulation ([Fefferman's Clay statement](#)).

Clay's September announcement describes the problem as "apparently" settled and says its evaluation process is deliberately unhurried, rather than announcing a completed institutional verification ([Clay Mathematics Institute announcement](#)). This review did not replay the external artifact or audit its entire formalization, and none of the counterexamples or algebraic corrections above depends on that artifact's acceptance.

Sentences requiring correction or a narrower label

The quotations below are from the supplied manuscript, not independently inspected Lean declarations. A valid kernel proof of a narrower statement would not validate the stronger prose.

Location and exact wording	Finding
Abstract: “finite iff the coarse description eventually fails; infinite exactly at the $\epsilon = 0$ pole where symbolic computation lives”	A sufficient error bound is not an if-and-only-if characterization of actual failure. Positive defect need not make every trajectory fail, and an exponential error estimate does not by itself give an unrestricted inverse-defect asymptotic.
Abstract: “finite iff the fine description continues”	The quoted abstract L0 theorem bounds a norm under a differential inequality; it does not supply an ODE/PDE extension theorem or an equivalence.
Section 3.2: “The Moore leg is the one SGC has actually formalized”	Requires source inspection. The named zero-defect tower theorem alone does not establish a generalized-shift Turing simulation or its continuum realization.
Section 3.1: “stationary Navier-Stokes states for any viscosity on manifolds with $H^1 \neq 0$ ”	Nontrivial first cohomology is not the sufficient hypothesis of the cited theorem: it requires a nowhere-vanishing harmonic carrier and a generally deformed metric, as discussed above.
Section 4.2: the SGC defect-budget row is “[KERNEL] at the abstract level (Theorem 2.7)”	Theorem 2.7 allows a density W ; identifying W with a closure defect requires an additional differential inequality. The non-claims elsewhere in the paper correctly admit that this bridge is missing.
Section 5: “the fine dynamics is stationary so no budget is spent”	A nonzero steady field accumulates $T \ \omega\ _\infty$ of BKM budget over time T . Stationarity does not make that density zero.
Section 6.1: “Its L^2 pairing with the fine field is exactly the energy flux across the cutoff N ”	False as written: the exact identity contains another term. A vector residual norm is also not a signed scalar flux.
Section 6.1: “every finite- N validity horizon closes before T^* ”	Does not follow from regularity loss. Fixed-cutoff defect divergence is excluded by the energy estimate above.
Section 6.1: “For a Beltrami field ($\text{curl } u = \lambda u$) ... $D_N(u) = 0$ for every N ”	Requires constant λ and a cutoff commuting with curl, or another proof that truncation preserves the Beltrami property. Full-field cancellation alone is insufficient; a manifold/contact realization is not automatically a flat Fourier realization.
Section 6.1: “Vorticity measures stretching”	Vorticity measures local rotation. Vortex stretching involves $(\omega \cdot \nabla)u$; strain and vorticity are not interchangeable.
Section 6.1: “their regularity criterion $\int_0^T \Lambda(t)^2 dt < \infty$ ”	For the Cheskidov–Shvydkoy dissipation cutoff, the cited L^2 criterion needs the additional Besov hypothesis described above. A different determining cutoff needs its own statement and proof.
Section 6.2: “Then for every N ... $\limsup \dots \ D_N\ = \infty$ ”	Incompatible with uniformly bounded energy at every fixed cutoff, including in any hypothetical finite-energy singularity.
Section 6.2: “Equivalently in determining-wavenumber terms”	No equivalence has been proved. A wavenumber criterion, a defect norm, and an actual approximation horizon are different quantities.
Section 6.2: “the computation, which requires $\epsilon = 0$ transparency across the scales it uses”	Necessity is not established by constructing one exact symbolic factor. Symbolic coding scales and physical Fourier scales have not been identified.

There is an additional elementary definition-cone check for the inaccessible CantorShiftTower source. Ordinary finite-window observation of the bilateral shift is not an autonomous deterministic factor: two sequences can agree on $[-n, n]$ and differ at $n + 1$, so their observed left shifts differ. The formal tower might instead use stochastic kernels, a different factor map,

different depths, or restricted states; its exact definitions must decide that question. This observation is a request for inspection, not an accusation that the kernel theorem is false.

Recommended decision gate

Do not commit a year to the present “flux controls Galerkin validity, and collapse destroys every fixed scale” formulation. A smaller, well-defined L1 program remains defensible, but its value is formal verification and potentially sharper estimates, not discovery of residual-based error control.

- **Falsification tests:** Formalize the $M = 3, N = 1$ counterexample and the fixed-cutoff energy bound before adding continuum claims. These should become regression tests for the paper’s modelling dictionary.
- **Correct interface:** Distinguish D_N, R_N, r_N , and Π_N in declarations and prose. Prove the exact error equation and flux identity rather than asserting a correspondence.
- **First positive estimate:** Prove the inhomogeneous energy/norm comparison for finite spectral systems, with the forcing tail and initial unresolved tail explicit. Then compare its statement and constants with the cited a posteriori literature.
- **Continuation target:** Choose the exact regularity density and reproduce its hypotheses. Do not skip from an L^2 error estimate to continuum smoothness or identify a determining cutoff with an approximation horizon.
- **Repository gate:** Finish the statement/receipt audit against an accessible immutable branch snapshot before marking any result independently verified. Keep mathematical novelty, kernel closure, model correspondence, and numerical validation as separate verdicts.

The useful research question is now: can SGC supply a rigorously defined, computable residual estimate or new structural control that improves on the existing a posteriori theory? Neither the two-horizon analogy nor zero-defect symbolic computation answers that question by itself.